

Product Requirements Document — AI Impact Evaluation v1.0

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Product Requirements Document — AI Impact Evaluation (v1.0)

- **Status:** v1.0 — Draft for stakeholder review (pending sign-off)
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- **Source of truth:** [AI_Impact_Evaluation_PRD_v1.0.docx](#) (this markdown mirrors it for day-to-day engineering use; the docx is authoritative for sign-off)
- **Upstream:** BRD v1.0 (see [brd-summary.md](#)). The BRD answers *why*; this PRD answers *what the product does, for whom, and to what acceptance standard*.
- **Downstream:** C4 architecture model ([../03-architecture/system-architecture.md](#)), metric formulas ([metric-definitions.md](#)), delivery backlog (Section 6 epics).
- **Delivery status against this PRD:** see [Appendix B](#).

1. Vision & product overview

Give every engineering leader a single, trustworthy, real-time answer to: **“How healthy is our software delivery, and what is our AI investment returning?”**

AI Impact Evaluation is a multi-tenant web platform. Teams connect their existing tools once; AI Impact Evaluation ingests, normalizes identities and team structures, computes DORA/SPACE-aligned metrics on a materialized semantic layer — no manual tagging, no spreadsheet rebuilds — and surfaces them as role-based drill-down dashboards, exportable executive reports, and (Phase 3) AI ROI and automated first-pass code review. It is an analytics layer over the toolchain, never a replacement for Jira/GitHub/GitLab.

Product goals

Goal	Description	BRD objective
PG-1	One source of truth for delivery performance across all teams	BO-1
PG-2	Fully automated DORA + SPACE-aligned metrics, no manual status updates	BO-2
PG-3	Quantify AI coding-assistant adoption and ROI in dollars	BO-3
PG-4	Reduce code-review cycle time and defect leakage	BO-4
PG-5	Make planned-vs-unplanned engineering time visible (Investment Profile)	BO-5
PG-6	Role-based views tuned to Exec, Manager, and IC needs	BO-6
PG-7	Commercialization-ready without re-architecture (optional path)	BO-7

Non-goals

Not a PM/Git-hosting replacement · no keystroke/idle-time/surveillance signals (deliberate product boundary, not a backlog gap) · no HR performance-review documents (deferred beyond Phase 3) · never requires engineers to change how they work.

Guiding product principles

1. **Trust over surveillance** — individual views opt-in and growth-framed; no metric may rank or shame an individual by default; metric definitions always visible.
2. **Insight, not raw data** — every screen answers a question a leader actually asks, with context, thresholds, and trend.
3. **Zero-tagging automation** — metrics derive from tool events; garbage-in-garbage-out is designed out.
4. **Fast time-to-value** — first meaningful dashboard within 30 minutes; no query language for core use.
5. **Accessible depth** — pre-built reports serve 90% of users; a guided custom-metric builder serves power users without SQL literacy.
6. **Resilient by design** — a vendor API outage degrades gracefully and never loses data or corrupts a metric.

2. Positioning

The deliberate middle ground: minware’s automated, no-tagging DORA/SPACE analytics delivered through Hivel’s leadership-friendly storytelling UX — explicitly avoiding surveillance (Hivel’s risk) and a mandatory query language (minware’s). Built internal-first but commercialization-ready: Phase 1 ships SaaS multi-tenant for internal pilots; the architecture must not preclude external onboarding, VPC/on-prem ingest, or SOC 2 Type II (see OQ-1).

3. Personas & jobs-to-be-done

Persona	Primary job-to-be-done	Primary surface	Access
CTO / VP Engineering	Report delivery speed, quality & AI ROI to the board with confidence	Executive Cockpit, AI ROI Summary	Org-wide
Engineering Manager	Find bottlenecks, balance workload, prep 1:1s from real data	Team Dashboard, Activity, Investment Profile	Team scope
Tech Lead / Senior Eng	Keep review load healthy and quality trends positive	Code Review Analytics	Team scope
Individual Contributor	See own trends transparently and non-punitively	Personal Activity (opt-in)	Self only
Product / Program Mgr	Predict delivery, see scope creep early	Investment Profile, Cycle-Time reports	Team scope
Finance / Procurement	Judge cost-benefit of AI tooling and AI Impact Evaluation itself	AI ROI Financial Report	Read-only, aggregated
Security / Compliance	Govern access, audit data use, enforce privacy	Admin & Access Console	Admin

Success moments (the moments that define product-market fit per persona): Exec exports a board-ready report with RAG DORA status and a dollar AI-ROI figure in under two minutes · Manager finds the sprint’s real bottleneck and drills to the stalled PRs without anyone being ranked · Tech Lead gets flagged on an aging oversized PR before it blocks release · IC opts in, sees clear definitions, trusts nobody is watching keystrokes · PM sees unplanned work rising before the milestone slips · Finance sees cost/quality/throughput deltas attributed to specific AI tooling · Compliance retrieves a 12-month audit log on demand.

4. Success metrics & instrumentation

North Star: weekly leaders acting on AI Impact Evaluation insight — count of Exec/Manager users taking a tracked action (drill-down, export, goal edit, follow-up on a flagged PR) at least once a week.

KPI	Target	How measured
Time-to-first-insight	< 30 min from first tool connection	connector_connected → first_dashboard_rendered
Manual reporting effort saved	70%+ reduction within 2 quarters	Manager survey + export usage
Team adoption	80%+ of target teams active within 90 days	Active-team instrumentation
Data-trust score	≥ 4/5 among engineering leads	In-product survey
AI ROI demonstrated	≥ 1 assistant with a \$ figure in first Phase-3 release	AI ROI module output
Dashboard load	< 3 s at ~1M events	Frontend performance telemetry
Metric freshness	≤ 15 min from source event	Ingestion → materialization lag metric

First-party instrumentation events (required from day one, aggregate-only, no surveillance content): connector_connected, first_dashboard_rendered, dashboard_viewed / drilldown_opened, report_exported, goal_created / goal_progress_viewed, pr_flag_actioned, ai_roi_report_viewed, survey_response_submitted.

5. Epic map & traceability

MoSCoW priority is noted on every story. Every story traces BRD → epic → phase → release.

Epic	Name	BRD source	Phase
E1	Onboarding & Connectors	FR-1.x, §5.1, §10	1
E2	Identity & Team Normalization	FR-1.4, FR-1.5, §11.1	1
E3	DORA & Delivery Metrics	§8.2, §5.1	1
E4	Cockpit / Executive Dashboard	§8.3, §5.1	1
E5	Investment Profile	§8.4, §5.1	2
E6	Code Review & PR Analytics	§8.5, §5.1	2
E7	Goals & OKR Tracking	§8.7, §5.2	2
E8	Administration & Access Control	§8.8, §9	1
E9	AI Adoption & ROI	§8.6, §5.2	3
E10	AI Code Review Agent	§8.5, §5.2	3
E11	Custom Reporting & Query Layer	§8.9, §5.2	3

6. Detailed requirements by epic

Acceptance criteria are the contract QA and engineering build to.

E1 • Onboarding & Connectors (Phase 1)

Goal: an admin connects the organization's tools in minutes over secure least-privilege scopes and reaches first insight in under 30 minutes. MVP connectors: Git host (GitHub and/or GitLab/Bitbucket), Jira, one CI/CD tool. Should-have: SonarQube, PagerDuty/Opsgenie, AI-assistant telemetry. Secrets encrypted; per-connector health surfaced to admins.

- **E1-S1 Connect a Git provider [Must]** — OAuth/app install requests only least-privilege read scopes; status shows Connected; historical backfill starts automatically with visible progress; auth failure/revocation shows an actionable error, never a silent failure; repos/orgs can be scoped in/out before ingestion.
- **E1-S2 Connect ticketing (Jira) [Must]** — epics, stories, sprints, status-change history ingest within the freshness target; projects map to AI Impact Evaluation teams during setup; non-standard workflows are tolerated and reported, not rejected.
- **E1-S3 Connect a CI/CD tool [Must]** — build/deploy events with timestamps and outcomes ingest reliably; deployments map to repos/services for DORA attribution; pipeline duration captured.
- **E1-S4 Resilient ingestion [Must]** — transient errors retried with backoff; persistent failures raise a connector-health alert; rate limits respected without dropping events (queue-and-drain); unexpected upstream schema changes are logged and quarantined rather than corrupting metrics.
- **E1-S5 30-minute time-to-value [Must]** — after first Git + ticketing connection, at least one populated dashboard renders with no manual configuration; guided setup checklist shows remaining steps and data-readiness; connect→first-render duration instrumented against the 30-minute target.

E2 · Identity & Team Normalization (Phase 1)

Goal: reconcile that one human is v.sharma in Git, Vishal S in Jira, and vishal@ in CI, and roll people up into teams — inaccurate identity resolution is the fastest way to lose leadership trust.

- **E2-S1 Contributor identity reconciliation [Must]** — multiple tool identities resolving to the same person attribute to a single canonical contributor; Admin can review/confirm/merge/split suggested matches; unresolved identities are flagged, never silently dropped or double-counted.
- **E2-S2 Team-structure import [Should]** — three-level org > team > sub-team hierarchy imports and is editable; reassignments recompute historical roll-ups consistently; teams without a source-of-record can be created manually.

E3 · DORA & Delivery Metrics (Phase 1)

Goal: the four DORA metrics plus lead-time breakdowns, automatically, zero manual tagging, filterable by team/repo/sprint/date. Formulas: [metric-definitions.md](#).

- **E3-S1 Automated DORA computation [Must]** — all four metrics compute with no manual tagging or status discipline; each shows current value, trend, and comparison to a configurable target band; every metric exposes a plain-language definition.
- **E3-S2 Lead-time stage breakdown [Must]** — commit → PR open → review → merge → deploy breakdown per team and repo; the slowest stage is visually emphasized; ticket-level lead time shown alongside PR-level.
- **E3-S3 Change Failure Rate from incidents [Should]** — deployments linked to a subsequent incident/hotfix/rollback count as failures; linkage logic is transparent and Admin-adjustable; MTTR derived from incident open/resolve timestamps.
- **E3-S4 Universal filtering [Must]** — team/repo/sprint/date filters apply consistently across all metric views and persist within a session; filtered results respect the viewer's access scope; empty results explain why, never a blank chart.

E4 · Cockpit / Executive Dashboard (Phase 1)

- **E4-S1 Single-pane Cockpit [Must]** — headline DORA + PR velocity + epic progress on one screen; RAG status per tile against configurable thresholds; loads within 3 s at

~1M events.

- **E4-S2 Org → team → IC drill-down [Must]** — drill-down preserves active filters and time range; individual-level drill-down gated by access control AND personal opt-in (E8); breadcrumbs navigate back up.
- **E4-S3 Presentation-ready export [Must]** — export produces a shareable formatted artifact (PDF/deck-ready) reflecting current filters; completes in seconds; logged as `report_exported`; metric definitions accompany the export.

E5 • Investment Profile (Phase 2)

- **E5-S1 Planned vs unplanned vs rework [Should]** — Git activity correlated to Jira epics classifies each unit of work; split shown per team over time with category definitions; unclassifiable work bucketed transparently.
- **E5-S2 Scope-creep trend [Should]** — rising unplanned-work trends visually surfaced with drill-down into contributing tickets.
- **E5-S3 Cost/time attribution for finance [Could]** — effort attributable to epics/initiatives with time-based cost estimates; methodology documented and exportable for audit.

E6 • Code Review & PR Analytics (Phase 2)

- **E6-S1 Review analytics [Should]** — PR size distribution, review turnaround, reviewer workload balance per team/repo; reviewer load shown as *balance*, never an individual leaderboard; respects access scope and privacy principles.
- **E6-S2 Proactive PR flags [Should]** — PRs exceeding configurable size/age thresholds flagged in-product; acting on a flag emits `pr_flag_actioned`; thresholds team-configurable with sensible defaults.

E7 • Goals & OKR Tracking (Phase 2)

- **E7-S1 Set metric-linked goals [Should]** — a goal binds to a metric, target value, scope, and time window; multiple concurrent goals per team; emits `goal_created`.
- **E7-S2 Automatic progress tracking [Should]** — progress recomputes as metrics refresh within the freshness target; on-track/at-risk/off-track shown against target trajectory; visible in the Cockpit for in-scope leaders.

E8 • Administration & Access Control (Phase 1)

Roles and default visibility: **Admin** (full config, connectors, roles, audit log) · **Engineering Leader** (org-wide aggregated + team drill-down) · **Manager** (own teams only) · **IC** (self only, opt-in personal view) · **Finance/Read-only** (aggregated ROI & cost views, no individual data).

- **E8-S1 Role-based access control [Must]** — the five roles enforce their default visibility; Admin can refine team/individual rules; no path (including exports and deep links) leaks data outside a user's scope; access changes take effect immediately and are logged.
- **E8-S2 Opt-in, non-surveillance individual views [Must]** — no keystroke, idle-time, or activity-surveillance metric exists anywhere; a manager cannot view an individual's personal view without the IC's opt-in per policy; every individual-level metric displays its definition in growth-oriented, non-ranking language.
- **E8-S3 Audit trail [Must]** — all configuration changes, access grants, and exports logged with actor, timestamp, target; retrievable ≥ 12 months; audit log itself access-controlled and tamper-evident.

E9 · AI Adoption & ROI (Phase 3)

- **E9-S1 Track AI-assisted work [Should]** — detection from assistant telemetry and/or commit conventions; confidence and method transparent; low-confidence attribution labeled; adoption per team over time.
- **E9-S2 AI vs non-AI delta [Should]** — cycle time, defect rate, and rework compared side by side with clear methodology; comparisons control for obvious confounders where feasible and state their limits.
- **E9-S3 Dollar ROI figure [Should]** — at least one assistant yields a \$ ROI figure in the first Phase-3 release; calculation assumptions visible and adjustable; high-value usage patterns and low-adoption pockets surfaced for enablement.

E10 · AI Code Review Agent (Phase 3)

- **E10-S1 First-pass automated review [Could]** — on PR open, the agent posts advisory comments (style, likely bugs, security), clearly attributed to the agent; humans remain the deciders; enable/disable per repository.
- **E10-S2 Review quality feedback loop [Could]** — before/after review-time and defect-leakage effects reported; false-positive rate tracked.

E11 · Custom Reporting & Query Layer (Phase 3)

- **E11-S1 Pre-built report library [Should]** — covers the most common use cases out of the box (cycle time, bug resolution, DORA workflow); filterable and exportable; no query language required.
- **E11-S2 Guided custom-metric builder [Could]** — custom metrics composed through a guided UI, not raw query text; saveable, shareable within scope, exportable; an advanced/raw mode may exist but is never required.

7. Key user flows

1. **First-run (Admin):** sign-in → guided setup checklist → connect Git (backfill with visible progress) → connect Jira + CI/CD, map to teams → identities normalized, unresolved matches flagged → first populated dashboard. *Success: populated DORA/Cockpit within 30 minutes, no query-writing, no engineer behavior change.*
2. **Board prep (Exec):** open Cockpit (RAG status) → drill amber Lead-Time tile org → team → quarter filter, confirm trend + linked goal → export presentation-ready report with definitions attached. *Success: board-ready report in under two minutes.*
3. **Sprint health (Manager/Tech Lead):** filter Team Dashboard to sprint → breakdown highlights review as slowest stage → open Code Review Analytics, stale oversized PR flagged → rebalance reviewer load, follow up (pr_flag_actioned). *Success: bottleneck actioned within the sprint; no individual ranked or shamed.*

8. Information architecture

Role-aware navigation — users see only surfaces their role and scope permit: **Cockpit** (Exec, Leader) · **Teams** (Manager, Leader) · **Code Review** (Tech Lead, Manager) · **Investment** (PM, Finance, Leader) · **Goals** (Leader, Manager) · **AI Insights** (Ph.3 — Exec, Finance, Leader) · **Reports** (Ph.3 — all analytical roles) · **Personal** (opt-in — IC) · **Admin** (Admin, Security).

9. UX & design requirements

Insight-first layout (answer first, raw detail one drill-down away) · definitions on demand inline for every metric · consistent RAG semantics against stated configurable targets · filters persist

and travel through drill-downs and exports within a session · growth-framed individual UX, never leaderboards · WCAG-conscious contrast and keyboard navigation; legible on laptop and large display · empty/loading states explain themselves (no data / out of range / still ingesting).

10. Non-functional requirements

Same as BRD §9 (see [brd-summary.md](#)): < 3 s dashboards at ~1M events · ≤ 15 min metric freshness · horizontal ingestion scaling, 10,000+ contributors · SOC 2 Type II readiness (P2/3), encryption in transit/at rest, least-privilege scopes · no surveillance metrics by design, opt-in individual views · 99.5% uptime, graceful ingestion degradation · < 30 min time-to-value · 12+ month audit retention · P1 SaaS multi-tenant, P3 VPC/on-prem ingest · new connectors without core re-architecture.

11. Data & privacy

Sources per BRD §10 (Git, Jira/PM, CI/CD, SonarQube, incidents, AI assistants, optional Slack/calendar), always least-privilege. **Privacy guarantees that must hold across all data:** no keystroke/idle-time/activity-surveillance data collected *or derivable* — full stop · individual views opt-in and access-gated, aggregate/team views default · metric definitions always transparent · data residency (VPC/on-prem ingest) may be needed earlier than Phase 3 for specific engagements.

12. Release plan

Phase	Duration	Epics	Exit criteria
0 — Discovery	2-3 wks	PRD sign-off, C4 model, pilot selection	BRD + PRD approved; pilots named; build-vs-buy quotes in
1 — MVP	8-10 wks	E1, E2, E3, E4, E8	Git+Jira+1 CI/CD live; DORA + Cockpit + RBAC; 30-min TTV met
2 — Depth	6-8 wks	E5, E6, E7 (+ code-quality connector)	Investment Profile, PR analytics, Goals shipped to pilot
3 — AI & Advanced	8-12 wks	E9, E10, E11 (+ VPC/on-prem option)	AI ROI \$ figure; custom reports; review agent piloted
4 — Commercialize (opt.)	Ongoing	Multi-tenant packaging, SOC 2 Type II	External onboarding ready (only if commercial path chosen)

MVP definition (Phase 1 exit): a pilot team connects Git + Jira + one CI/CD tool, reaches a populated DORA Cockpit in under 30 minutes, drills org→team with RBAC enforced, and exports a leadership-ready report — zero manual tagging, zero surveillance metrics.

13. Dependencies, assumptions, constraints, out of scope

Dependencies: third-party API access/scopes (blocks all metrics) · pilot team availability · engineering sponsorship for adoption · AI-assistant telemetry or commit conventions (weakens E9 attribution if absent) · C4 model from Technical Architect · build-vs-buy decision. **Assumptions/constraints/out-of-scope:** per BRD §12/§5.3 — supported Git host + ticketing in place; 2-3 pilot teams; build-vs-buy budget not yet approved; vendor rate limits constrain refresh; surveillance permanently excluded; no tool replacement; HR review automation deferred.

14. Open questions (must close Phase 0)

ID	Question	Owner	Blocks
OQ-1	Internal tool, commercial SaaS, or both?	Sponsor + Product	Phase 4 scope, pricing
OQ-2	Build, buy, or hybrid vs Hivel/minware?	Finance/Procurement	Build scope & budget
OQ-3	Which 2-3 teams are the Phase-1 pilot?	Sponsor	Phase 1 start
OQ-4	Which single CI/CD tool ships first in MVP?	Technical Architect	E1, E3 scope
OQ-5	Pricing model if commercialized (flat-team recommended)?	Product + Finance	Phase 4
OQ-6	Is VPC/on-prem needed before Phase 3 for any engagement?	Security + Sales	Deployment architecture
OQ-7	What is the org's individual-view opt-in policy?	People + Security	E8 personal views

Appendix A · Glossary

Product-facing definitions live in the PRD docx Appendix A and must appear inline in-product wherever the corresponding metric is shown. Engineering-grade formulas, data sources, and edge cases live in [metric-definitions.md](#) — that file is the computational source of truth.

Appendix B · Delivery status

Repo-only appendix (not in the signed docx) — updated as increments land. Legend: [Done] done · [Partial] partial · [Not started] not started.

Epic / story	Status	Notes
E1-S1 Git provider	[Partial]	connector-github: signature-verified webhooks + PR/commit backfill live; GitHub App install flow, repo scoping UI, GitLab pending
E1-S2 Jira	[Partial]	connector-jira: token-verified webhooks + issue backfill with changelogs; project→team mapping UI pending
E1-S3 CI/CD	[Partial]	GitHub Actions via connector-github (workflow-run backfill + live events); deployment-rule mapping pending (OQ-4 answered as GitHub Actions for pilot)
E1-S4 Resilient ingestion	[Partial]	Queue + retry + DLQ + idempotent staging writes live and smoke-verified; GET /api/v1/admin/connectors (ADMIN) now surfaces per-connector status/last- sync/event-count derived from staging.raw_event, rendered live in the Admin console; proactive alerting (paging) on top of this signal still pending
E1-S5 30-min TTV	[Partial]	GET /api/v1/setup/status (ADMIN/ENG_LEADER) derives a 4-item checklist (git/ticketing/CI/dashboard) and a firstConnec- tionAt→firstDashboardAt time-to-value figure entirely from stag- ing.raw_event/mart.metric_daily timestamps — no manual flags; frontend Setup view renders it. Guided remediation steps (e.g. deep links to reconnect a failed connector) pending

Epic / story	Status	Notes
E2-S1 Identity reconciliation	[Partial]	identity-service: alias resolution, exact-email merge (confidence-scored), bot detection, git-signature fallback; Admin review/merge/split UI pending
E2-S2 Team import	[Partial]	connector-github fetches org teams + repos + members; identity-service upserts core.team/core.team_repo/core.team_member resolving members through the existing identity resolver; sub-team hierarchy (parent_team_id) not yet populated
E3-S1 DORA metrics	[Partial]	metrics-engine + mart: all four DORA metrics live end-to-end — deployment frequency (DORA-1), lead time (DORA-2, heuristic PR-open→deploy), change failure rate (DORA-3), MTTR (DORA-4) via hotfix/rollback linkage; plus PR velocity/cycle-time — each at repo, org, and team scope. RAG target bands and incident-based CFR (Phase 2) pending
E3-S2 Lead-time stages	[Partial]	Total lead time + deploy-wait computable; commit→PR and review-wait/review stages need PR-review + commit-in-PR ingestion
E3-S4 Filtering	[Partial]	Repo/org/team scope + date-window filters live in the API (scope param); sprint dimension pending
E4-S1 Cockpit	[Partial]	Real tiles from /api/v1/metrics/cockpit with freshness indicator, loading/error/empty states, definitions on hover; RAG thresholds and epic progress pending

Epic / story	Status	Notes
E4-S2 Org → team drill-down	[Partial]	GET /api/v1/teams + Cockpit scope param let the frontend list teams and drill into team-scoped tiles (smoke-verified); IC-level drill-down and breadcrumb-preserved filters beyond team pending
E8-S1 RBAC	[Partial]	api-core enforces five roles server-side (JWT resource server, ADR-0004); metrics + teams endpoints gated to analytical roles, IC denied; dev-token bridge pending OIDC; Admin console now renders live connector health and audit log (frontend Admin.tsx wired to /api/v1/admin/connectors + /api/v1/audit), replacing two of its three mock panels; role-assignment management still mock (no persisted user directory yet — needs OIDC/user-management backend) and per-user team/individual visibility rules pending
E8-S2 Opt-in individual views	[Partial]	Invariant enforced (no surveillance metrics exist; IC denied org/team metrics); personal-view endpoint + opt-in gate pending its surface
E8-S3 Audit trail	[Partial]	Append-only audit log with admin-only read; token issuance audited; config/export events wire in as those surfaces land; cryptographic tamper-evidence is later hardening
E5-E7, E9-E11	[Not started]	Phase 2/3